2024 2nd DMIHER International Conference on Artificial Intelligence in Healthcare, Education and Industry (IDICAIEI) | 979-8-3315-2871-3/24/$31.00 ©2024 IEEE | DOI: 10.1109/IDICAIEI61867.2024.10842736

Development of an Efficient Supervised Machine Learning Model using Concept Drift Analysis: A Review

Abhay A. Dande
Computer Science and Engineering Department
PRMIT&R, Badnera, 444701
Amravati (M.H.), India
dandeabhay@gmail.com

Dr. Mahendra A. Pund
Computer Science and Engineering Department
PRMIT&R, Badnera,444701
Amravati (M.H.), India
pundmukesh@gmail.com

***Abstract*— The need of efficiently working supervised machine learning models is of great importance and it is the domain of widespread research in the top Information Technology sector industry. Presently machine learning models are becoming very complex, and it is necessary to maintain the efficiency of the models in the increased complexity for better performance. Efficiency of the complex supervised machine learning models plays a significant role in the improved user experience, cost and resource optimization, real time applications, model interpretability, explainability and its reuse ability. Organizations are able to utilize the full capability of the supervised learning models and can have the expected outcome in the allied sector with efficient supervised machine leaning models. Machine learning is the part of the almost any industry and has its effect on many aspects of human life. To design the expected workable and efficient machine learning model is really challenging. It mainly includes the importance steps such as to select the proper data sets, accurate feature analysis, selecting the proper machine learning model type, architecture, hyper parameters finally designing efficient model. Concept drift is the major challenge in this design process and that needs to be analyzed with systematic approach. This review paper aims to discuss Concept drift analysis with respect to supervised machine learning models.**

Keywords*— *Concept Drift, Precision, Sensitivity, Specificity, F1 Score, Accuracy, Mean Squared Error

I. Introduction

Machine Learning is one of the subfields of artificial intelligence that deals with the design of algorithms, methodologies and models that incorporate the ability to computers and machines, to learn from the data, understand and make predictions from the obtained knowledge without any need of the explicit programming. In recent years machine learning has witnessed considerable growth and proliferation with the advancement in computing ability, availability of the large standard datasets and innovative research contributions by the experts in this area. The concept of the machine learning is under study from many decades but due to advancement in the machine learning algorithms and more specialized domains in the machine learning such as deep learning, natural language processing and large language models it has been the most popular technology in almost all the industry sectors. The ability of the machine learning models to analyse the vast amount of data and to get the valuable conclusions has made it an important tool for researchers and business.

II. Literature Review

Authors J. Lu et.al [1] explains the review of about 130 good quality scientific publications in the area of the concept drift analysis. It has the up-to-date analysis of the algorithms, techniques to handle the concept drift and suggests a framework type approach for concept drift analysis. The paper has a well-defined list of artificial datasets and benchmark datasets that can be used to analyse the performance efficiency of the machine learning models in dealing with the concept drift. It also mentions the details for the concept drift related core concepts.

Authors N. Hatamikhah et.al [2]. This research paper deals with the concept drift analysis using the concept of deep neural network. The research approach mentioned in this paper is based on ensemble algorithm and "streaming ensemble algorithm" (SEA) algorithm. It has the focus on the parameter of observed change in the error that occurs in classification with respect to the newly arrived data. It has given more importance to the functioning of the deep learning algorithm.

Authors Pingfan Wang et.al [3] proposed a model centered framework for the precise detection of the concept drift using the concepts of deep learning and transfer learning. It focuses more on the change in the model itself instead of change in the model output. The transfer learning approach used in this paper speed the concept drift analysis process and reduces the incorrect identification of the concept drift. It uses the standard artificial databases and also real-world application databases to demonstrate the usefulness of the proposed approach.

Authors Firas Bayram et.al [4] disclose the sound review of the various concept drift detection methods invented in recent years. These methods mostly make use of the degradation in the predictive ability of the model as a major indicator for change in data distribution and related concept drift detection and its further analysis. It has also mentioned the importance of the hyper parameter tuning for the efficient machine learning model design.

Authors Hanqing Hu et.al [5] focused on concept drift detection using ensemble framework approach. It uses the combination of multiple set of classifiers for concept drift detection. The conclusion results obtained from these algorithms are effectively summarized using a novel mechanism called voting by type. The experimentation mentioned in the research paper is carried out on artificial datasets and three real world data sets.

Authors Samuel Ackerman et.al [6] deals with the concept drift detection in the supervised machine learning model using

XXX-X-XXXX-XXXX-X/XX/$XX.00 ©20XX IEEE

Authorized licensed use limited to: Universiti Pertahanan Nasional Malaysia. Downloaded on October 18,2026 at 20:00:02 UTC from IEEE Xplore. Restrictions apply.

concept of weak data slices. The method proposed in this paper makes the use of feature space rule which is referred as data slice in case of concept drift detection. This approach is able to detect the performance degradation in the model for concept drift using novel approach that works even without the actual feature labels.

Authors P. Siirtola et.al [7] deals with the importance of features in the datasets to analyse the concept drift. It takes into consideration the entire number of features and their correlation. It also takes into account the total number of records for each type feature and their quantitative analysis. The observed shift in the correlation of the features denotes the presence of the concept drift. The data distribution is then processed to analyse and handle the concept drift.

Authors E. Roffe et.al [8] suggests a novel approach to detect the concept drift in machine learning models using polynomial relations. The change in the data distribution is detected with the consideration of polynomial relations between the various features of the dataset. R square value indicates the strength of the polynomial relations between the features of the data set. The high values of the R square indicates that these are the important traits of the dataset that are supposed to remain constant if there is no significant change in the data distribution. Fast Switch Naive Bayes to Avoid Redundant Update for Concept Drift Learning by A.

Liu, G. Zhang et.al [9] deals with analysis of the false indications of the concept drift using Naive Bayes method. It explains the concept that the performance parameter of the supervised machine learning model is highly sensitive to nature and amount of training data. These research papers provide valuable insights for the precise analysis of concept drift and to study its impact on the efficiency of the supervised machine learning models. The following table 1 shows the efficient concept draft analysis methods.

TABLE I. EFFICIENT CONCEPT DRAFT ANALYSIS METHODS

Author	**Analysis parameter**		***Limitations***
	Method	***Advantages***	
J. Lu et al. (2020)	Framework-based approach, benchmark datasets	Comprehensive analysis, well-defined core concepts	High complexity, extensive computational resources required
N. Hatamikhah et al. (2021)	Deep neural network, ensemble algorithm	Improved classification accuracy	Requires large training datasets
Pingfan Wang et al. (2022)	Model-centered framework, transfer learning	Reduces incorrect drift detection, speeds up analysis	Dependent on pre-trained models
Firas Bayram et al. (2023)	Performance-aware drift detectors	Focus on predictive performance degradation	Limited to specific use-cases only
Hanqing Hu et al. (2023)	Ensemble framework, voting by type	Effective summarization of results	High Complexity in implementing ensemble methods

III. IDENTIFIED CORE OBJECTIVES FOR CONCEPT DRAFT ANALYSIS

Analysis of the concept drift is a major challenge in designing efficient supervised machine learning models. Concept drift plays vital role in the degradation of the predictive ability of the model and lowers its performance. The research domain area Development of an Efficient Supervised Machine Learning Model with an Emphasis on Concept Drift Analysis for Labelled Data deals with the following major aspects.

A. Designing an efficient supervised machine learning model

The term efficiency of the supervised machine learning model mainly deals with the accurate predictive ability of the model. This predictive ability of the model should not degrade even in case of the newly unseen datasets and even in data distribution change. The efficiency of the model should not get reduced in case of increased complexity of the model and its related hyper parameters. It should work sufficiently well for large scale application of wide domain [10, 11].

B. The challenge of concept drift analysis

This research proposal deals with the challenge of analysing the concept drift in the machine learning models. This consists of the detection, analysis and adaption of the model to handle this concept drift. Concept drifts deal with the change in the statistical properties of data and data distribution. Many different approaches such as ensemble learning, polynomial relations and data slices approach are proposed until now to handle the concept drift [12].

C. Improving model adaptability

The efficient machine learning model should have adaptability for the occurrence of the concept drift, change in the data distributions and change in the statistical properties of the data. The reviewed research proposals possibly address effective handling of the concept drift and model adaption to maintain the efficiency and predictive ability of the model [13].

D. Analysing model performance and explainability

This paper need to analyse the performance of the model in relation to the concept drift. The important efficiency parameters and the evaluation metrics needs to be properly checked. Along with the accurate predictive ability of the model, the contribution of every feature in this prediction needs to be analysed so that the behaviour of the model can also be understood and trusted by its user [14]. Overall, the research problems and reviews focus around developing an efficient supervised machine learning model that will have the ability to detect, analyse and adapt to the concept drift while maintaining the accuracy.

This paper considers the research objectives aligned with the domain of development of efficient supervised machine learning models with focus on concept drift analysis. This paper summarizes these review objectives as follows.

E. Analyse the existing models

Analyse the working of existing supervised machine learning models including the models that shows the performance degradation over time. Investigate the positive and negative aspects of these models along with their strength and

Authorized licensed use limited to: Universiti Pertahanan Nasional Malaysia. Downloaded on October 18,2026 at 20:00:02 UTC from IEEE Xplore. Restrictions apply.

weakness. See the impact of the new datasets on the predictive ability of these models [15].

F. Identify the concept drift detection mechanisms

Study and understand various concept drift detection techniques for supervised machine learning models. Analyse the concept drift parameters, its impact and corrective measure to reduce the effect of this concept drift. Find the best possible mechanism that will considerably reduce the impact of the concept drift [16].

G. Develop the efficient supervised models

Design and develop efficient supervised machine learning model based on the identified efficiency parameters. The mechanism implemented to reduce the effect of the concept drift must contribute to the increased efficiency and adaptable performance for the predictive ability of the model [17].

H. Design an experimental prototype model:

The designed efficient supervised machine learning model can be tested for any real-world use cases applications based on the appropriate available datasets.

IV. Discussed Approach

This paper considers various applicable, and dataset related statistical properties of the data such as mean, median, mode, frequency, standard deviation, variance, range, skewness, kurtosis, outliers, correlation, and dimensionality of the data to detect the concept drift in the dataset with respect to the features in the dataset [18].

The data distribution of the data needs to be analysed to study the concept drift. The various types of data distribution tests that can be applied depends up the nature of the dataset. Some of the standard data distributions that can be used are Gaussian Distribution, Nominal Distribution, Poisson Distribution, Exponential Distribution, Binomial Distribution, Multimodal Distribution. If required these distributions can be used along with some good statistical tests such as Kolmogorov-Smirnov (KS) Test and Chi-Square Test [19, 20].

Initially this paper propose here the method based on the feature relevance approach for initial concept drift analysis. It deals with the identified set of features that are having largest relevance and contribution for the predictive ability of the model. To find out the feature correlation this paper proposed to use Pearson's correlation, Heatmaps can be efficiently used to find out the most correlated and most important features in the model's predictive ability. When there is a change in the statistical properties of the data, change in the data distribution leading to concept drift, this relevance between the features and the relevance between the predictive ability of the model will change in a notable manner indicating inefficiency in the model thereby degrading its predictive performance. This change in the relevance because of the occurred concept drift is crucial to monitor with the newly arrived data. In this case of the concept drift the feature relevance metric has to keep updated using incremental learning approach [1, 21, 22].

Once the baseline for feature relevance has been established it can be used as a reference point for the detection of the concept drift. The obtained current relevance every time with newly arrived data will be compared the baseline by the implemented concept drift detection method. The significant deviation from the baseline denotes the occurrence of the concept drift. Later on, model can be adapted accordingly to handle this deviation with the update in the model architecture and model parameters. The information obtained from the feature relevance can be used to guide the process of feature selection on which feature to prioritize or which feature to deprioritize in the concept drift processing of the model [23, 24].

The above-mentioned feature relevance approach gives more importance to feature and their statistical parameters. This paper exploring here is the combination of the deep learning algorithms and statistical concepts for feature correlation matrix and observed concept drift. The use of deep learning approach has the focus on the model design itself in occurrence of the concept drift rather than the output of the model. It reduces the complexity of the computation to a considerable extent and reduce the number of the wrongly identified concept drift and is comparatively takes very less time [2, 25].

The mentioned proposed approach for concept drift identification and adaption for efficient machine learning model design can be applied and explained in general as shown in the figure below.

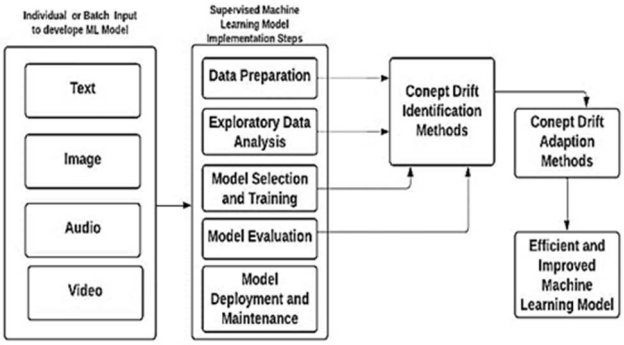


Fig 1: Concept Drift Analysis Process

The proposed model centric transfer learning coupled with statistical analysis and deep learning algorithms is quite superior in many respects. This approach speeds up the concept drift detection to a considerable extent reducing the complexity and computation cost. It is able to filter out the wrongly identified concept drift with great precision and accuracy. It allows the model to adapt according to the occurred concept drift. This adaption is reflected from the changes in a model structure such as feature weight change in deep neural network with back propagation and change in the decision boundary[26, 27].

V. Proposed Parameters to Analyze the Efficiency of Machine Learning Models

A. Accuracy

Accuracy of the machine learning model is the primary metric and it is the measure of the total number of correct predictions instances by the model out of all the available instance for a particular dataset. It is appropriate measure for the balanced datasets but for the unbalanced datasets it does not work well.

B. Precision

Authorized licensed use limited to: Universiti Pertahanan Nasional Malaysia. Downloaded on October 18,2026 at 20:00:02 UTC from IEEE Xplore. Restrictions apply.

This parameter measures the accuracy of the positively predicted instance made by the model. It is defined as the ratio of the identified positively predicted instance to the total number of positive instances present.

C. Recall or Sensitivity

Recall measures the ability of the model to correctly identify all positive instances present. It is also called as the true positive rate.

D. Specificity

This parameter measures the ability of the model to correctly identify all negative instances present. It is also called as true negative rate

E. F1-Score

The F1-score is the harmonic mean of precision and recall. It is used to balance precision and recall. It is the useful metrics for efficiency if the dataset is imbalanced.

F. ROC Curve and AUC

Receiver Operating Characteristic (ROC) and Area Under the ROC Curve (AUC). It is the graph of the parameters true positive rate and true negative rate at various identified threshold values. It is used to quantify the overall performance of the model. High value of the AUC indicates good quality of discrimination between classes.

G. Mean Square Error (MSE)

It is the performance metric used in case of the regression models. It is the measurement of the average of the squared difference between the values predicted by the model and the actual values of the target variable.

H. Root Mean Square Error (RMSE)

RSME is obtained from the MSE. It is the square root of the MSE. It is generally used when the error metric needs to in the same units as the target variable.

I. Absolute Error (AE)

It is also called as Mean Absolute Error (MAE). It is not much sensitive to outliers in comparison with MSE and RMSE. It is the measurement of the absolute differences between the values predicted by the model and the actual values of the target variable.

J. Confusion Matrix

It is a metric used for the determination of the performance of the classification model for a given dataset. It gives the information about the nature of the predictions made by the model. It has the entries classified as True Positive, True Negative, False Positive, and False Negative.

VI. CONCLUSION

With the successful analysis of the concept drift, the model can be auto adjusted to maintain its desired performance that ensures its possible expected efficiency and desired predictive ability for the real worlds use cases. Early alarm system guides the time-to-time interventions lowering the effect on performance of the model and preventing potential damage due to wrong interpretations. With the continuous monitoring of the data the machine learning models incorporate themselves with the auto update ability and the generalization capabilities of the models gets enhanced making them more suitable for scaling ability. The users are able to get deep insight into model's behaviour and provides confidence in the model's decision-making ability. Instead of unnecessary training the model on regular basis the models get retrained on the occurrence of concept drift only. This approach helps to reduce the complexity, computational cost and saves time.

REFERENCES

[1] J. Lu, A. Lu, F. Dong, G. Gu, J. Gama, and G. Zhang, "Learning under Concept Drift: A Review," *arXiv:2004.05785v1* [cs.LG], Apr. 13, 2020.

[2] N. Hatamikhah, M. Barai, M. R. Kangavari, and M. A. Keyvanrad, "Concept Drift Detection via Improved Deep Belief Network," in *2018 IEEE Xplore*, pp. 978-1-5386-4916-9/18, 2018.

[3] P. Wang, N. Jin, D. Davies, and W. L. Woo, "Model Centric Transfer Learning for Concept Drift Detection," *Knowledge-Based Systems*, vol. 275, p. 110705, 2023.

[4] F. Bayram, B. S. Ahmed, and A. Kassler, "From concept drift to model degradation: An overview on performance-aware drift detectors," *Knowledge-Based Systems*, vol. 245, p. 108632, Jun. 7, 2022.

[5] H. Hu, M. Kantardzic, and L. Lyu, "Detecting Different Types of Concept Drifts With Ensemble Framework," in *2018 17th IEEE International Conference on Machine Learning and Applications*, 2018, pp. 978-1-5386-6805-4/18.

[6] S. Ackerman, P. Dube, E. Farchi, O. Raz, and M. Zalmanovici, "Machine Learning Model Drift Detection Via Weak Data Slices," *arXiv:2108.05319v1* [cs.LG], Aug. 11, 2021.

[7] P. Siirtola and J. Roning, "Feature Relevance Analysis to Explain Concept Drift - A Case Study in Human Activity Recognition," in *Proceedings of the 2022 ACM International Joint Conference on Pervasive and Ubiquitous Computing and the 2022 ACM International Symposium on Wearable Computers*, Sep. 2022, pp. 386-391.

[8] E. Roffe, S. Ackerman, O. Raz, and E. Farchi, "Detecting model drift using polynomial relations," *arXiv:2110.12506v2* [cs.LG], Dec. 22, 2021.

[9] A. Liu, G. Zhang, K. Wang, and J. Lu, "Fast Switch Naive Bayes to Avoid Redundant Update for Concept Drift Learning," in *IEEE Xplore*, 2020, pp. 978-1-7281-6926-2/20.

[10] A. Tsymbal, M. Pechenizkiy, P. Cunningham, and S. Puuronen, "Dynamic integration of classifiers for handling concept drift," *Information Fusion*, vol. 9, no. 1, pp. 56-68, 2008.

[11] J. Gama, A. Bifet, M. Pechenizkiy, and A. Bouchachia, "A survey on concept drift adaptation," *ACM Computing Surveys (CSUR)*, vol. 46, no. 4, pp. 1-37, 2014.

[12] G. Widmer and M. Kubat, "Learning in the presence of concept drift and hidden contexts," *Machine Learning*, vol. 23, no. 1, pp. 69-101, 1996.

[13] J. Lee, "Detection of Concept Drift for Learning from Stream Data," in *IEEE 14th International Conference on High Performance Computing and Communications*, IEEE Computer Society, 2012.

[14] S. Yu, Z. Abraham, and H. Wang, "Concept Drift Detection and Adaptation with Hierarchical Hypothesis Testing," *Journal of the Franklin Institute*, vol. 356, no. 5, pp. 3187-3215, Mar. 2019.

[15] B. Kurlej and M. Wozniak, "Active learning approach to concept drift problem," in *Oxford University Press*, Oct. 2011.

[16] B. Krawczyk, "Reacting to Different Types of Concept Drift with Adaptive and Incremental One-Class Classifiers," in *IEEE Computer Society Publication*, 2015.

[17] D. Shang and G. Zhang, "Fast Concept Drift Detection Using Singular Vector Decomposition," in *IEEE 12th International Conference on Intelligent Systems and Knowledge Engineering (ISKE)*, 2017.

[18] B. Zhang and Y. Chen, "Research on detection and integration classification based on concept drift of data stream," *SpringerOpen Journal*, May 2019.

[19] G. Fenza and C. Stanzione, "Concept-drift detection index based on fuzzy formal concept analysis for fake news classifiers," *Technological Forecasting & Social Change*, vol. 194, p. 122640, May 2023.

Authorized licensed use limited to: Universiti Pertahanan Nasional Malaysia. Downloaded on October 18,2026 at 20:00:02 UTC from IEEE Xplore. Restrictions apply.

[20] B. Zhang and L. Xue, "Model Updating Mechanism of Concept Drift Detection in Data Stream Based on Classifier Pool," *SpringerOpen Journal*, May 2016.

[21] D. Janzing and L. Minorics, "Feature relevance quantification in explainable AI: A causal problem," *arXiv:1910.13413v2* [stat.ML], Nov. 27, 2019.

[22] L. Pfannschmidt and J. Jakob, "Feature Relevance Determination for Ordinal Regression in the Context of Feature Redundancies and Privileged Information," *arXiv:1912.04832v1* [cs.LG], Dec. 10, 2019.

[23] A. Liu and J. Lu, "Concept Drift Detection: Dealing with Missing Values via Fuzzy Distance Estimations," *arXiv:2008.03662v1* [cs.LG], Aug. 9, 2020.

[24] C. Halstead and Y. S. Koh, "Analyzing and repairing concept drift adaptation in data stream classification," *SpringerOpen Journal*, May 2021.

[25] J. Liu and J. Lu, "Concept Drift Detection via Equal Intensity k-means Space Partitioning," *arXiv:2004.11587v1* [cs.LG], Apr. 24, 2020.

[26] S. Sun and K. Tang, "Concept Drift Adaptation by Exploiting Historical Knowledge," *arXiv:1702.03500v1* [cs.LG], Feb. 12, 2017.

[27] D. Zambon and C. Alippi, "Concept Drift and Anomaly Detection in Graph Streams," *arXiv:1706.06941v3* [cs.LG], Feb. 9, 2018.

Authorized licensed use limited to: Universiti Pertahanan Nasional Malaysia. Downloaded on October 18,2026 at 20:00:02 UTC from IEEE Xplore. Restrictions apply.